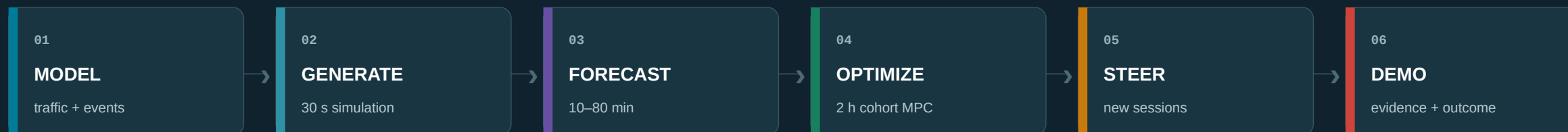


Predictive UPF Steering Simulation to Working Demo

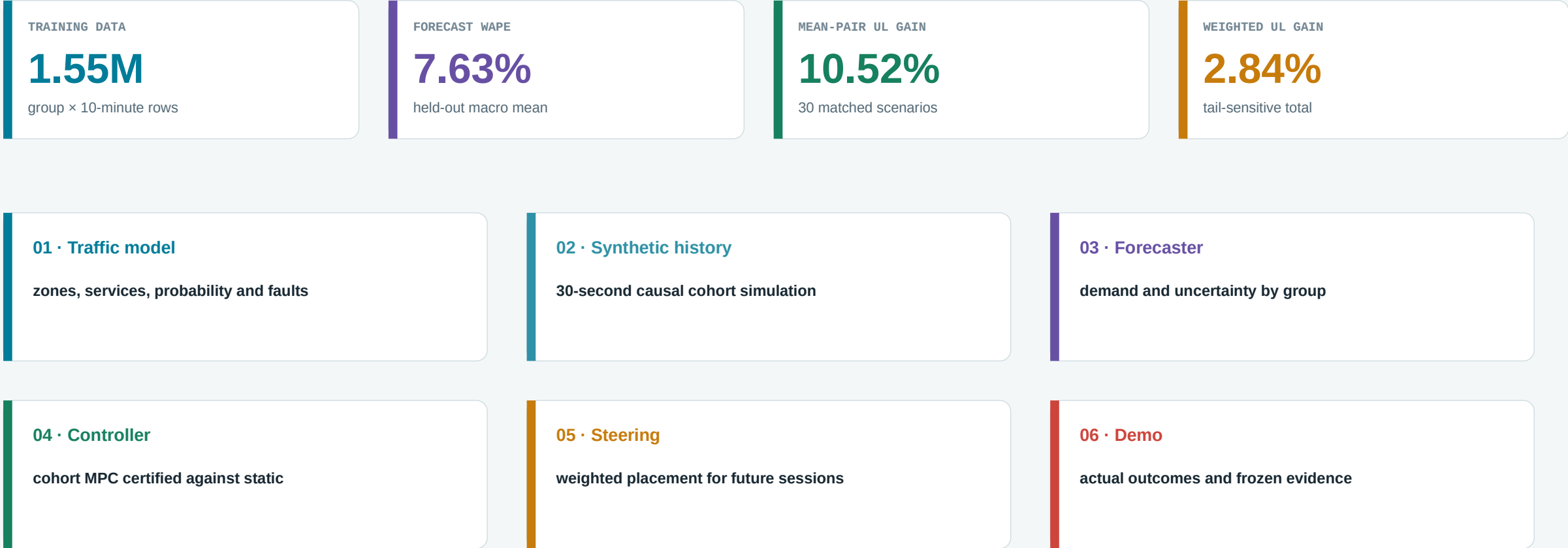
Data modelling, forecasting, controller experiments and evidence

Concise technical review · 09 August 2026



Synthetic, reproducible and explicitly outside a production-release claim.

A complete predictive steering loop is implemented and measurable



Claim: reduced modeled overload exposure for future sessions—not guaranteed prevention, migration or live C-DOT actuation.

Stage 1 — model diverse traffic as controllable groups

Dimensions

ZONES

8

urban · airport · stadium · industrial · rural

GROUPS

96

zone × DNN × slice × 5QI

12 SERVICE CLASSES

- Consumer video
- Gaming
- Enterprise
- Industrial URLLC
- Connected vehicle
- Cloud backup
- Social/live
- IMS voice
- Video conference
- Massive IoT
- Edge AI
- Public safety

Executable probability model

$$N(g, t) \sim \text{Poisson}(\lambda_g \times F_{g, t})$$

$F_{g, t}$ = daily curve × weekend factor × weekly noise × surge multiplier

- independent SHA-derived arrival and lifetime streams
- uniform integer session lifetime by class
- fixed class-level UL/DL Mbps per active session
- sessions retained as anchored cohorts after placement
- 5QI is forecast/QoS metadata, not a separate v1 steering key

Stage 1 — generate normal regimes, stochastic surges and network stress

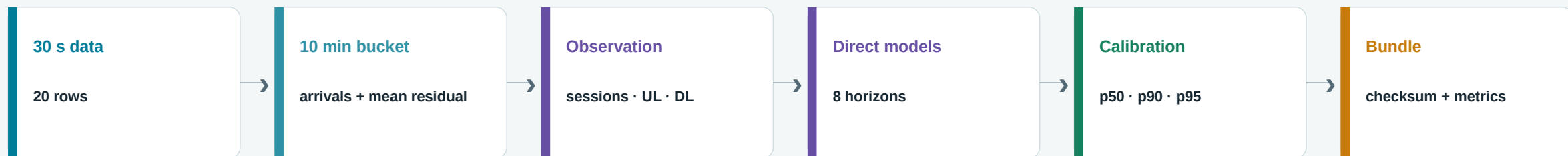
DAILY / WEEKLY	258,048 factors	service curves + weekend + U(0.86,1.16)
FLASH CROWDS	192 episodes	12/week · random group · 2–10 h · ×2.5–8
BROWNOUTS	128 episodes	8/week · UL capacity ×0.18–0.70
NEAR OUTAGES	48 episodes	3/week · degraded + 1% capacity
LATENCY	80 episodes	5/week · +25–140 ms

Boundary: no mobility, persistent UE identity, heavy-tailed rates, packets or generated telemetry faults in the extreme profile.

16-week corpus	
112 days	322,560 ticks
16,128	10-minute buckets
1,548,288	group observations
24 UPFs	16 edge · 4 regional · 4 central
≈4.39 B	projected session arrivals
≈9.6 GiB	projected artifact volume

Outputs: run.parquet, group/UPF buckets, selection audits and hashed metadata.

Stage 2 — transform the history into causal demand forecasts



Model

Calendar-ridge direct multi-horizon

- 96 groups × 3 targets × 8 horizons = 2,304 models
- last, MA(6), trend, daily lag and calendar Fourier features
- 70% train · 15% calibration · 15% chronological test
- median bias correction + split conformal uncertainty

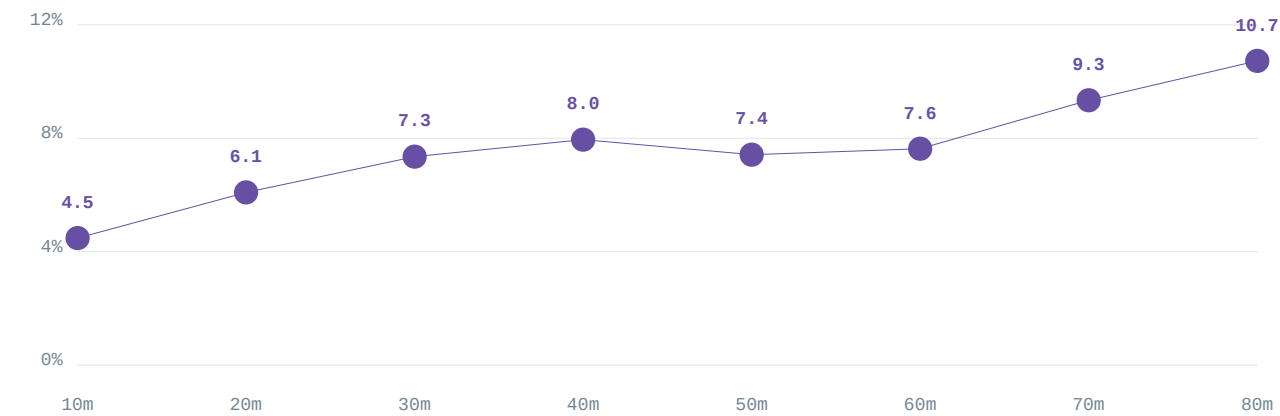
Training optimizations

- vectorized $O(n)$ feature construction
- RMS scaling for national-scale numerical stability
- immutable JSON bundle with internal SHA-256
- adaptive conformal alpha after realized coverage

3:03 total · ~13 s fitting · ~1.8 GiB RSS

Stage 2 — forecasting beats causal baselines, but surges remain difficult

Held-out WAPE by horizon



Overall p90 coverage 94.21% · p95 coverage 96.69%

Baseline comparison



44.36% WAPE reduction vs seasonal naive

NORMAL

6.51% WAPE

94.19% p90

SURGE

11.08% WAPE

30.73% p90

BROWNOUT

5.09% WAPE

87.88% p90

OUTAGE

5.29% WAPE

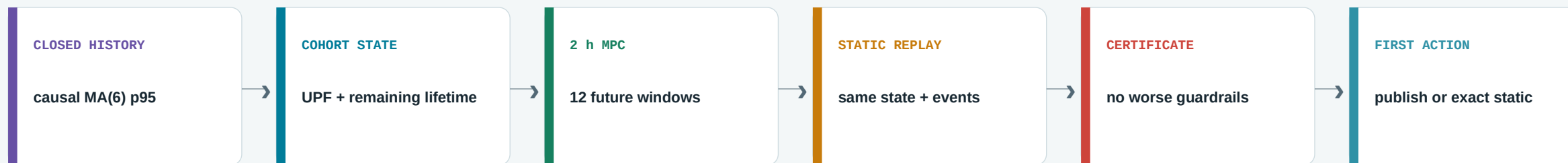
73.96% p90

Stage 3 — experiments showed why a one-window optimizer was not enough

EXPERIMENT	WHAT CHANGED	RESULT / DECISION
One-day predictive pilot	trained p95 forecast + one-window HiGHS LP	+2.40% UL area; insufficient for campaign
Eight-profile tuning	gate, costs, safety factors and static blends	all validation profiles lost to static
Mechanism isolation	scheduled hints, anomaly, lifetime and concentration cap	lifetime helped, but no profile passed
Oracle action-space bound	full-day cohort survival + perfect fault knowledge	100% modeled UL reduction was feasible
Cohort MPC development	two-hour state transition + static certificate	~25.09% mean on five dev seeds
First 12-pair pilot	broader surge/fault/outage/mixed matrix	failed: 4.74% mean; profile rejected
Frozen MA6 MPC	50% static blend + unknown-fault fallback	advanced to fresh 30-pair campaign

Lesson: placements persist for hours; optimizing only the next bucket can create residual concentrations that later decisions cannot undo.

Stage 3 — the accepted controller plans cohorts, certifies the action and steers new sessions



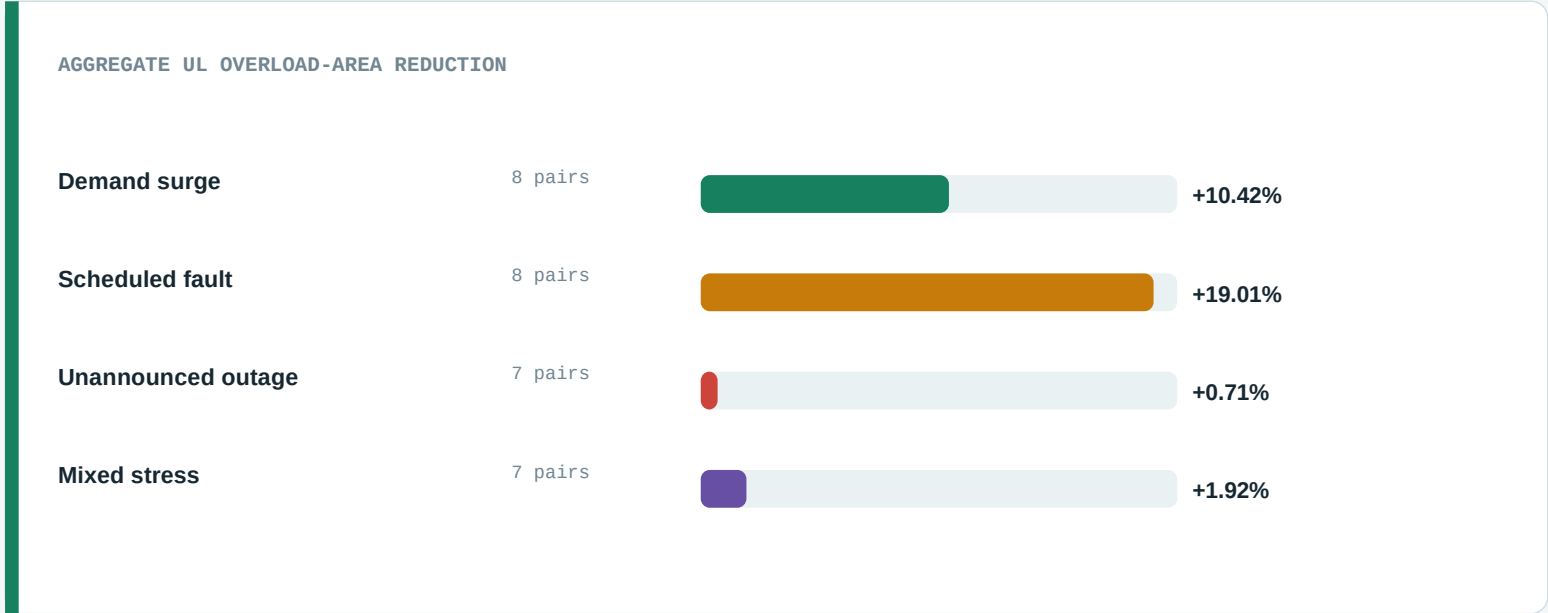
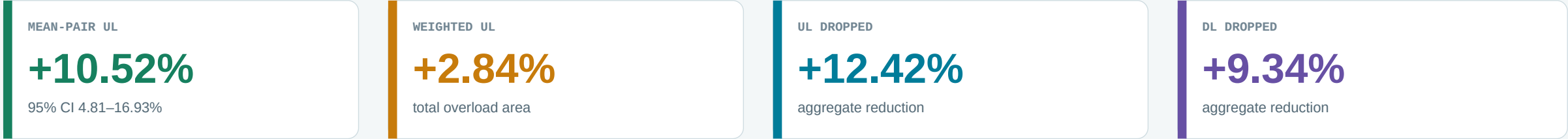
Frozen MPC profile

- 12 × 10-minute receding horizon
- exact anchored cohort state and expected survival
- known future capacity path only after known_at_step
- 50% optimized action + 50% contemporaneous static
- fallback to static during unplanned capacity state
- UL, DL, drop and session guardrails

How steering works

- MPC outputs normalized weights per traffic group and eligible UPF.
- Unhealthy or ineligible destinations are removed and weights renormalized.
- Each arriving session uses deterministic weighted rendezvous hashing.
- Admission checks the selected UPF session limit.
- The session stays anchored until its lifetime ends—no migration.

Stage 3 — the 30-pair campaign passes the demo gate, with visible tail risk

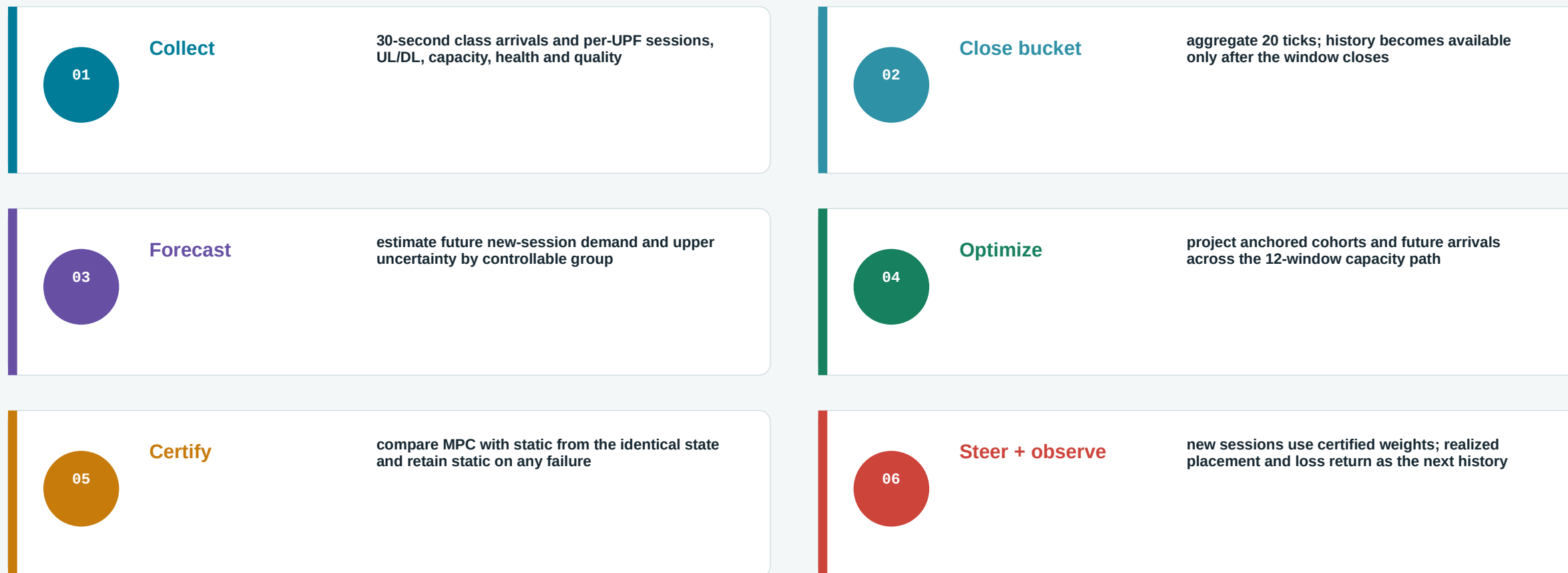


Tail and release boundary

Worst pair: -23.50%

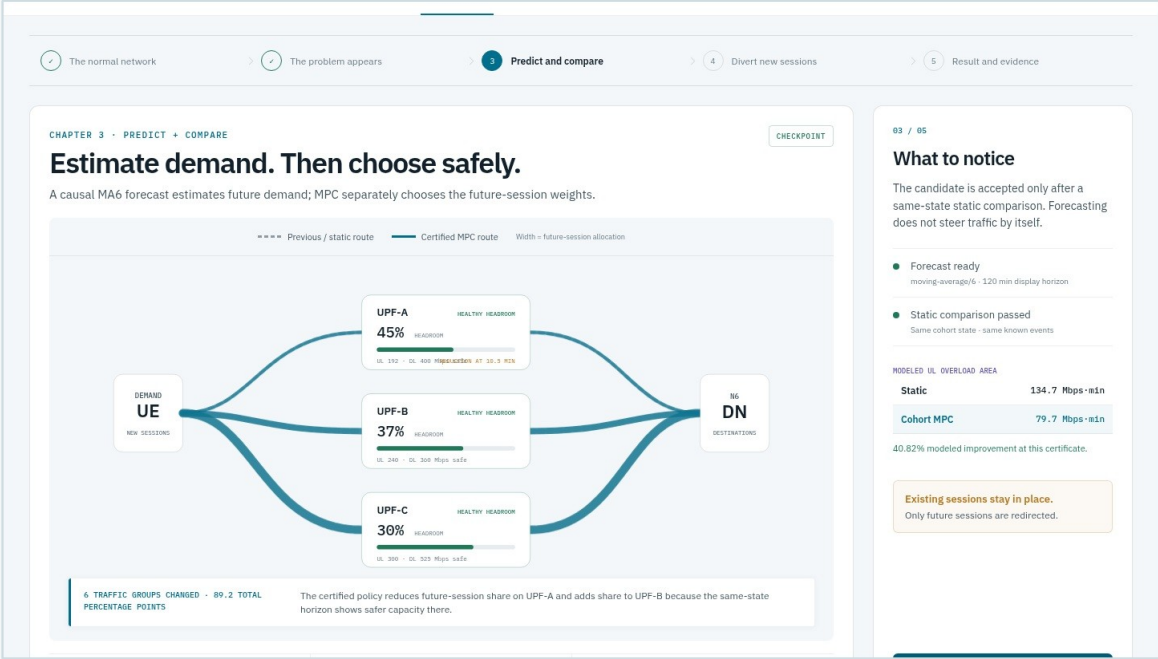
- all aggregate directional guardrails pass
- fault-heavy individual regressions remain
- working demo candidate—not production-ready

From incoming data to a steering decision: one causal 10-minute iteration



The dashboard displays this same loop; it does not substitute a prerecorded result for the live causal state.

The demo focuses on traffic pressure, steering and evidence



A short guided scenario explains what changes—and what does not

STEP	AUDIENCE SEES	INTERPRETATION
1 · Normal	all UPFs inside safe envelopes	existing sessions are anchored
2 · Pressure	class demand rises; capacity threat becomes visible	forecast inputs are still causal
3 · Predict	future demand + same-state static comparison	forecast estimates; MPC chooses
4 · Divert	future-session route widths change	weights change, not existing sessions
5 · Result	actual placement, error, loss and frozen campaign	reduced exposure—not perfect prevention

Key language: ‘future sessions redirected’ and ‘modeled exposure reduced’; never ‘traffic migrated’ or ‘overload prevented.’

What is ready today—and what C-DOT must provide for a calibrated pilot

Ready in the repository

- deterministic multi-class traffic and fault simulation
- canonical artifacts and reproducible experiment evaluation
- forecast bundle with uncertainty and causal inference
- static-certified cohort MPC and safe fallback
- new-session-only simulated steering
- working evidence-oriented dashboard

Needed from C-DOT / testbed

- live telemetry names, labels, quality and counter semantics
- measured directional and session capacity envelopes
- real topology, locality and UPF eligibility rules
- authenticated SMF/EMS policy publication interface
- decision on established-session migration capability
- shadow evaluation, untouched release seeds and tail gate

Recommended sequence: map telemetry → calibrate capacities → advisory replay → shadow recommendations → bounded new-session pilot.

The evidence supports a credible working demo and a clear next pilot

**Model the traffic.
Forecast the pressure.
Certify before steering.**

The strongest result is not one headline number—it is an end-to-end loop whose assumptions, experiments, gains and limitations remain traceable.

EVIDENCE SNAPSHOT

- 112-day synthetic history
- 7.63% forecast WAPE
- 10.52% mean-pair UL gain
- 2.84% severity-weighted gain
- all aggregate guardrails pass
- fault-tail robustness still open
- new-session steering only

Proposed discussion

Which C-DOT telemetry and actuation interfaces can support a calibrated shadow pilot?